MOLECULAR GENETICS

# Module map

- Module 07: Molecular Genetics
- Connected lab: lab-08_molecular-genetics.md
- Use this deck as the visual path through the module's evidence and retrieval work.

## Module 07
Molecular Genetics

1. DNA structure
2. Replication
3. Transcription
4. Translation
5. Mutation and gene expression evidence

**Lab evidence**
lab-08_molecular-genetics.md

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# Learning objectives

- Describe DNA structure and base-pairing rules.
- Explain why DNA replication is semiconservative.
- Compare transcription and translation.
- Use codons to connect mRNA sequence to amino acid sequence.
- Explain how mutation can alter protein and trait outcomes.

**OBJECTIVE 1**
Describe DNA structure and base-pairing rules.

**OBJECTIVE 2**
Explain why DNA replication is semiconservative.

**OBJECTIVE 3**
Compare transcription and translation.

**OBJECTIVE 4**
Use codons to connect mRNA sequence to amino acid sequence.

**OBJECTIVE 5**
Explain how mutation can alter protein and trait outcomes.

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# Topic sequence

- DNA structure: DNA stores information in nucleotide sequences and complementary base pairing.
- Replication: Replication copies DNA using each strand as a template.
- Transcription: Transcription produces RNA from a DNA gene.
- Translation: Translation uses mRNA codons and ribosomes to build proteins.
- Mutation and gene expression evidence: Mutations and expression patterns connect molecular changes to traits.

01

### DNA structure

DNA stores information in nucleotide sequences and complementary base pairing.

02

### Replication

Replication copies DNA using each strand as a template.

03

### Transcription

Transcription produces RNA from a DNA gene.

04

### Translation

Translation uses mRNA codons and ribosomes to build proteins.

05

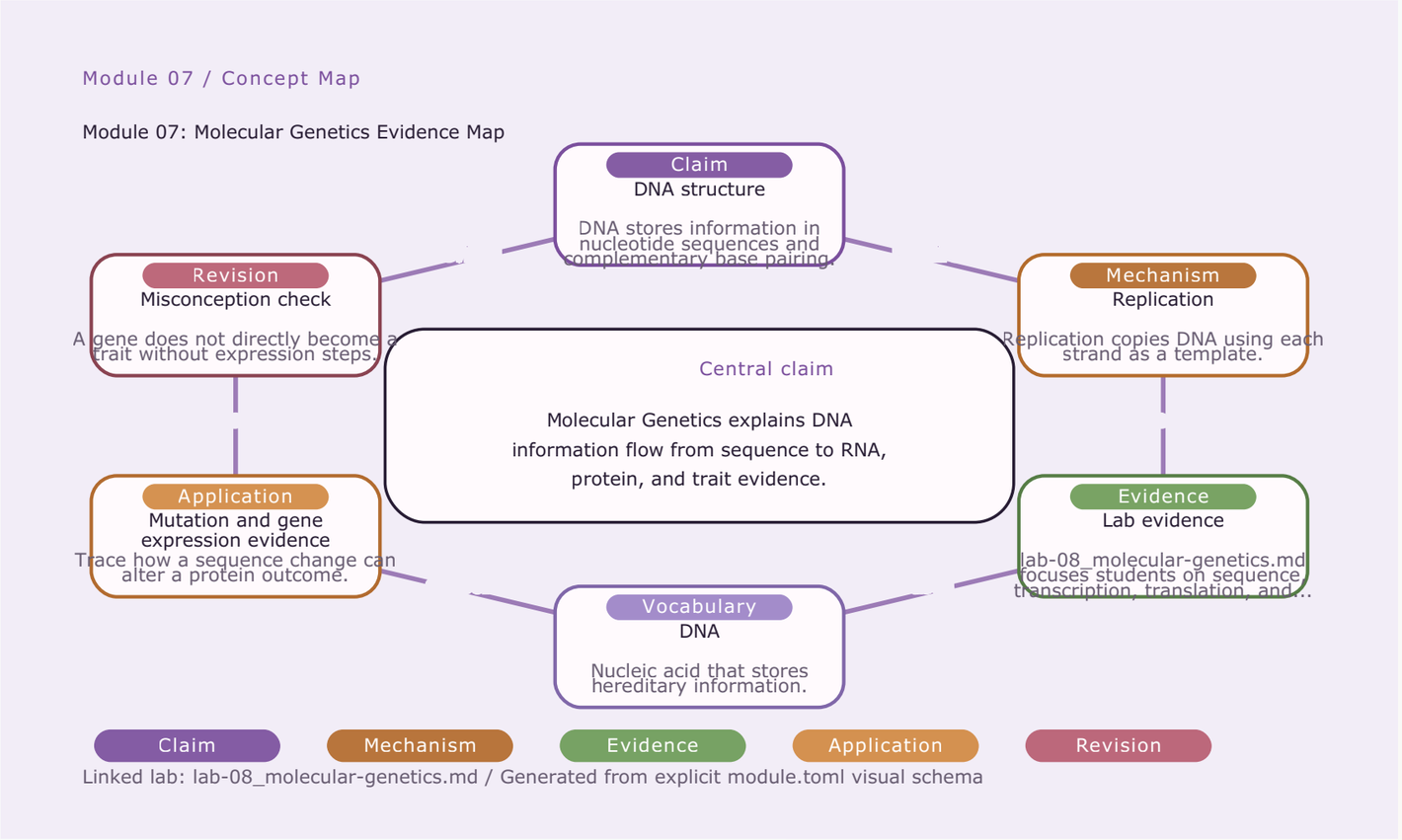
### Mutation and gene expression evidence

Mutations and expression patterns connect molecular changes to traits.

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# Module 07: Molecular Genetics Evidence Map

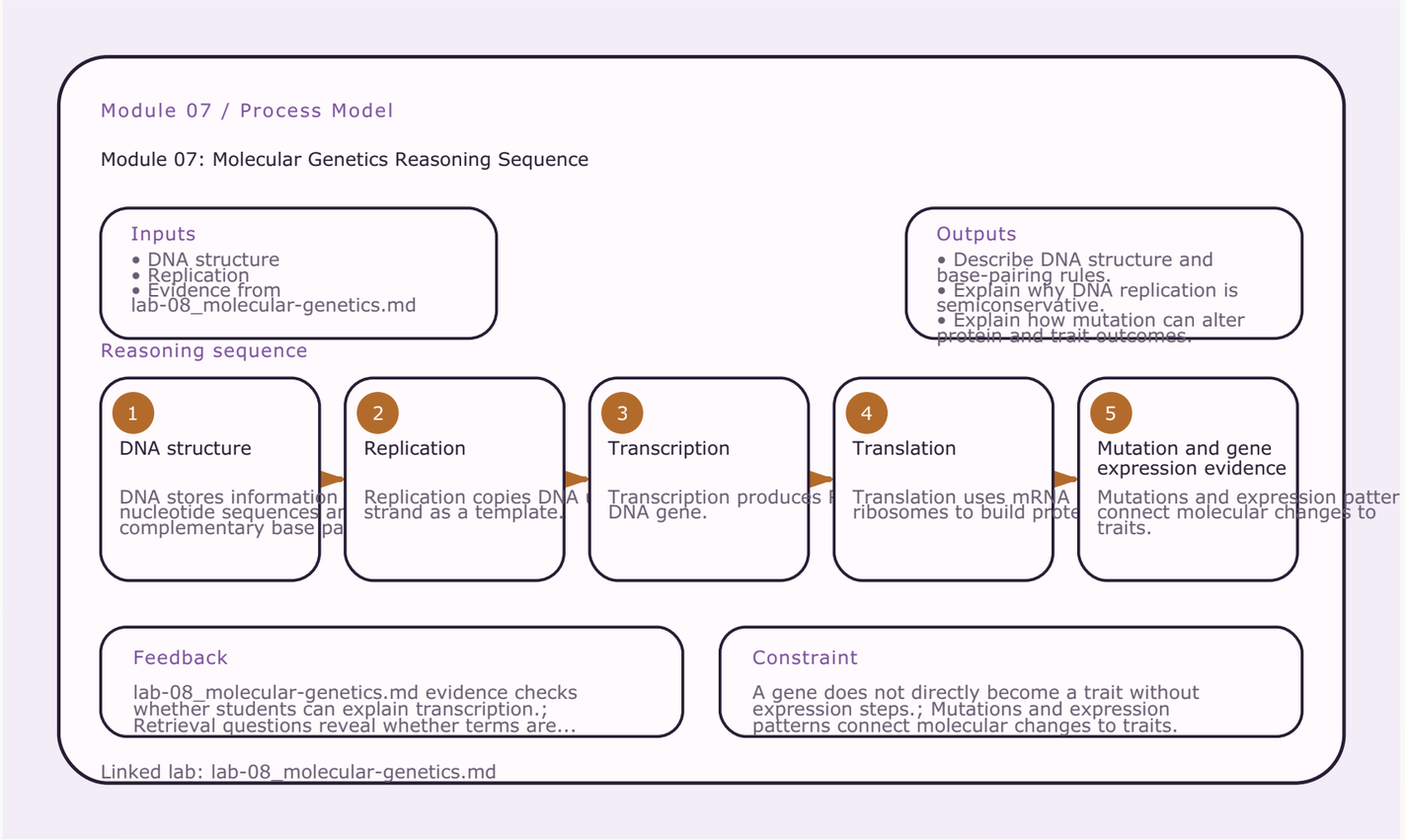
- Module focus: Molecular Genetics
- Central claim: Molecular Genetics explains DNA information flow from sequence to RNA, protein, and trait evidence.
- Key nodes to connect: DNA structure, Replication, Lab evidence
- Reasoning move: use 'requires' to explain relationships, not memorize terms.
- Lab connection: lab-08_molecular-genetics.md supplies an evidence surface for the map.



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# Module 07: Molecular Genetics Reasoning Sequence

- Module focus: Molecular Genetics
- Trace the model: DNA structure -> Replication -> Transcription -> Translation
- Inputs become outputs: DNA structure, Replication -> Describe DNA structure and base-pairing rules., Explain why DNA replication is semiconservative.
- Feedback check: lab-08_molecular-genetics.md evidence checks whether students can explain transcription.
- Lab connection: lab-08_molecular-genetics.md gives students a process to observe or test.



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# Terms as evidence handles

- DNA: Nucleic acid that stores hereditary information.
- Nucleotide: DNA or RNA subunit with sugar, phosphate, and base.
- Replication: Process that copies DNA.
- Transcription: Synthesis of RNA from a DNA template.
- Translation: Synthesis of protein from mRNA codons.
- Codon: Three-base mRNA sequence read during translation.

**DNA**
Nucleic acid that stores hereditary information.

**NUCLEOTIDE**
DNA or RNA subunit with sugar, phosphate, and base.

**REPLICATION**
Process that copies DNA.

**TRANSCRIPTION**
Synthesis of RNA from a DNA template.

**TRANSLATION**
Synthesis of protein from mRNA codons.

**CODON**
Three-base mRNA sequence read during translation.

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# Lab connection

- Lab file: lab-08_molecular-genetics.md
- Lab evidence should test or illustrate: Transcription
- Students should leave with a concrete observation, comparison, or model tied to the module claim.

**QUESTION**
DNA structure

**EVIDENCE**
lab-08_molecular-genetics.md

**CLAIM**
Describe DNA structure and base-pairing rules.

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# Contrast check

- Do not stop at: DNA stores information in nucleotide sequences and complementary base pairing.
- Stronger explanation: Mutations and expression patterns connect molecular changes to traits.
- The goal is to move from a named idea to a testable biological explanation.

**SURFACE**

DNA stores information in nucleotide sequences and complementary base pairing.

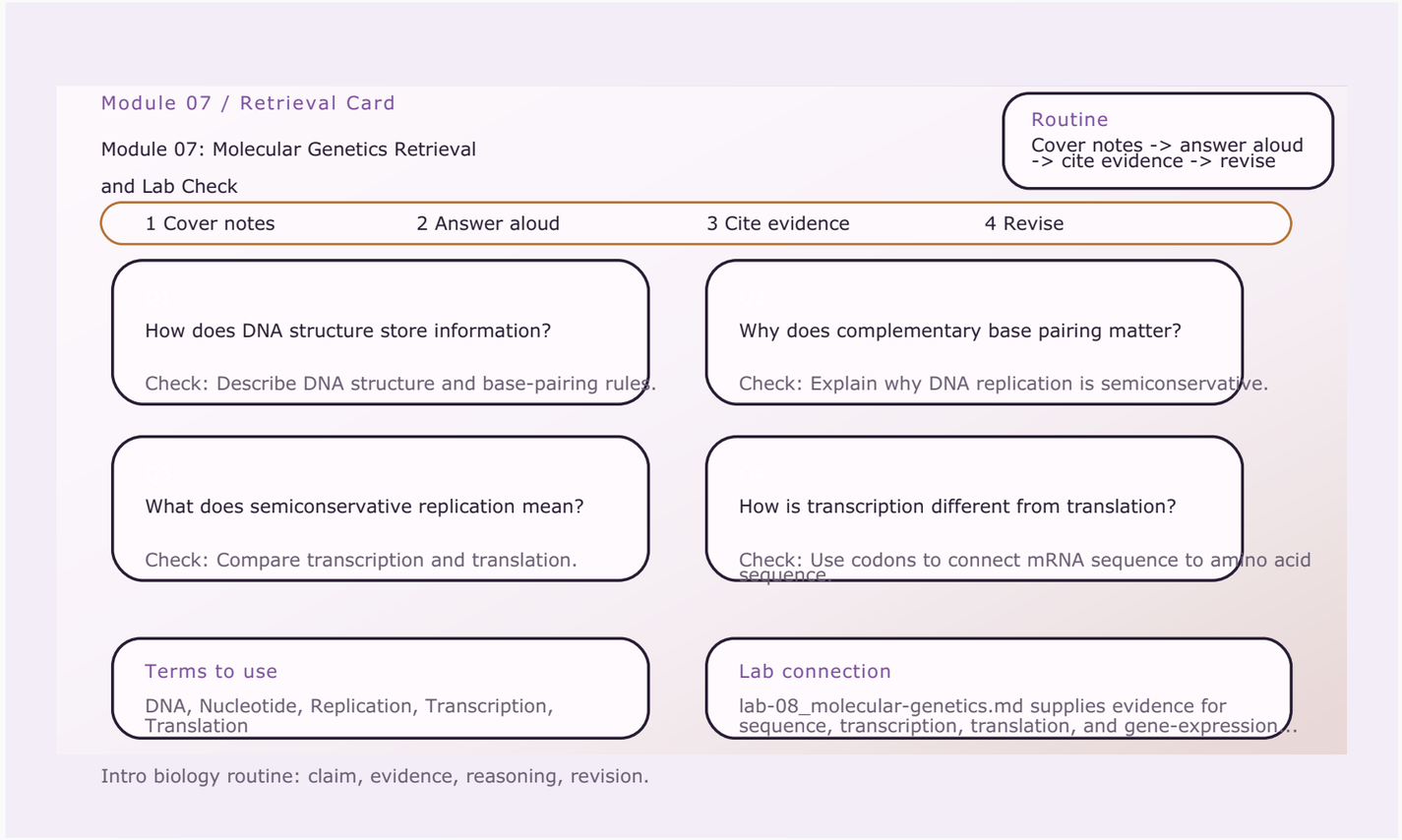
**DEEPER**

Mutations and expression patterns connect molecular changes to traits.

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# Module 07: Molecular Genetics Retrieval and Lab Check

- Module focus: Molecular Genetics
- Retrieval prompt: How does DNA structure store information?
- Answer check: Describe DNA structure and base-pairing rules.
- Required terms: DNA, Nucleotide, Replication
- Lab connection: lab-08_molecular-genetics.md; lab-08_molecular-genetics.md supplies evidence for sequence, transcription, translation, and gene-expression evidence.



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# Practice quiz bridge

- 1. In DNA, the base adenine always pairs with:
- 2. DNA replication is described as "semiconservative" because each new double helix contains:
- 3. Which sequence shows the normal flow of genetic information?
- 4. A mutation changes a single DNA base. Why might this matter?

**QUESTION 1**

## Answer first

In DNA, the base adenine always pairs with:

**QUESTION 2**

## Answer first

DNA replication is described as "semiconservative" because each new double helix contains:

**QUESTION 3**

## Answer first

Which sequence shows the normal flow of genetic information?

**QUESTION 4**

## Answer first

A mutation changes a single DNA base. Why might this matter?

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# Synthesis and exit ticket

- Exit claim: explain dna structure using evidence from lab-08_molecular-genetics.md.
- Must include: DNA, Nucleotide, Replication.
- Revision prompt: what evidence would change your explanation?

**CLAIM**
DNA structure

**EVIDENCE**
lab-08_molecular-genetics.md

**REVISION**
What would change your mind?